

Linear systems

Question 1 / 10 points / Blank assignment

Solve this system using row reduction. Show and label each row operation.

Write the final answer as a column vector.

$$2x + y = 5$$

$$x - y = 1$$

Proof by induction

Question 2 / 10 points / Blank assignment

Prove the following statement for every integer $n \geq 1$ by induction.

$$1 + 3 + \dots + (2n - 1) = n^2$$

State the inductive hypothesis for arbitrary $k \geq 1$ and finish with an explicit conclusion that the statement holds for every integer $n \geq 1$.